

E-Commerce Sales Analysis

Python-based Data Cleaning, EDA and Visualization

Executive Summary

This project analyzed an e-commerce sales dataset using Python, Pandas, Matplotlib and Seaborn. The analysis covered data cleaning, exploratory data analysis (EDA), univariate and categorical analysis, monthly trends, relationships between numerical variables, and business-focused visualizations. The objective was to identify revenue patterns, customer and product behavior, operational opportunities, and areas where the business can make more data-driven decisions.

Revenue & Product Performance

1. **Electronics** is the Best Selling product & generate the most Revenue while **Beauty** contribute the least.
2. Revenue is positively skewed, with most transactions concentrated at relatively lower revenue values and a smaller number of transactions generating substantially higher revenue. The revenue boxplot also revealed multiple potential high-value outliers.

Discount and Revenue Relationship

The Discount Vs Revenue scatter plot shows a large amount of variation in revenue at almost every discount level. There is no strong positive relationship between discount and revenue, while the visual pattern may indicate a weak negative tendency.

Discount alone does not appear to be a strong explanation for revenue differences.

Quantity and Revenue

The Quantity Vs Revenue scatter plot indicates that higher quantities generally tend to be associated with higher revenue. However, substantial variation remains, showing that quantity is not the only driver of transaction value.

- Large-quantity purchases can contribute strongly to revenue.
- Product price and product mix also influence revenue, so quantity should be evaluated together with pricing information.
- Bundles, cross-selling and quantity-based offers may be useful strategies for increasing basket size, provided profitability is maintained.

Product Category Performance

Product categories were evaluated using total revenue, quantity sold, average order value and average customer rating. This multi-dimensional approach is more useful than judging categories by total revenue alone.

- A category with high total revenue may be successful because it has a large number of orders.
- A category with fewer orders but a higher average order value can still be strategically important.
- Category-level customer ratings can help identify categories requiring product-quality or service improvements.

Regional Performance

Regional analysis compared total revenue, average revenue and average delivery days. The results provide a combined view of financial performance and operational efficiency.

- Regions with strong revenue but relatively long delivery times may represent opportunities for logistics optimization.
- Lower-performing regions can be investigated for differences in demand, product mix, pricing, customer behavior and availability.
- Regional KPIs should be monitored together rather than relying on revenue alone.

Payment Method Behavior

Payment methods were compared based on order frequency. This identifies the payment options most commonly used by customers.

- The most-used payment methods should remain reliable and convenient.
- Less-used payment methods can be evaluated to determine whether limited adoption is caused by customer preference, availability or usability.
- Payment-method performance can be combined with revenue and customer segments in future analysis.

Monthly Business Trends

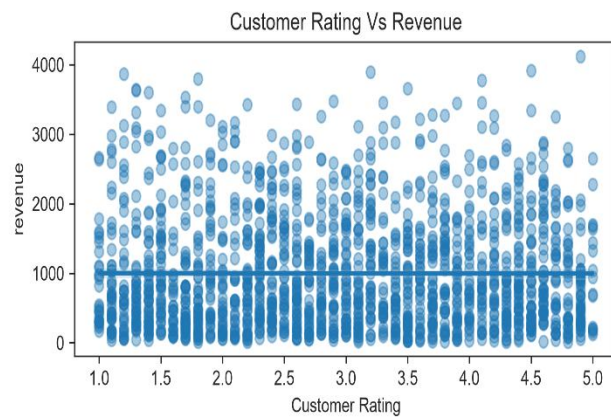
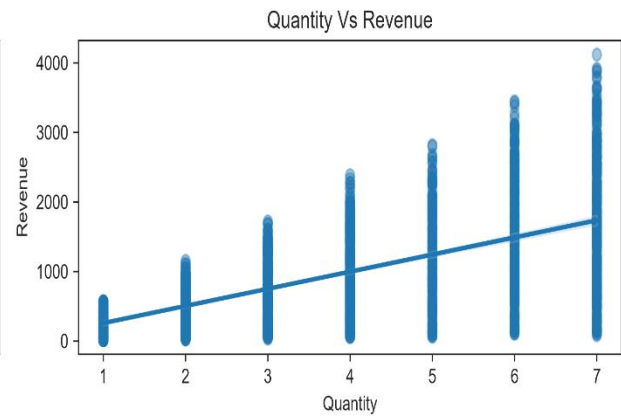
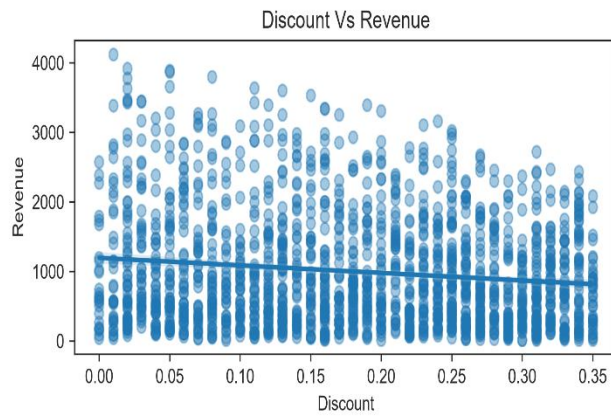
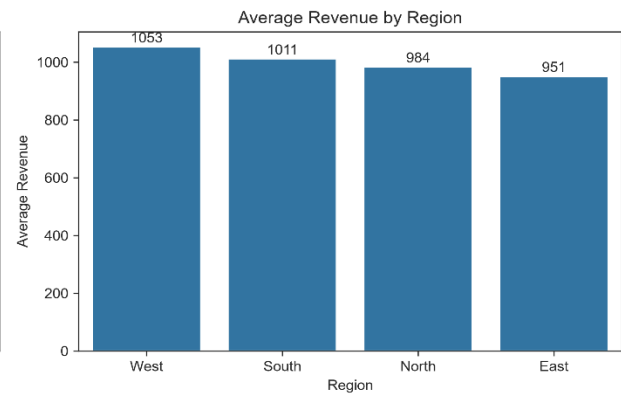
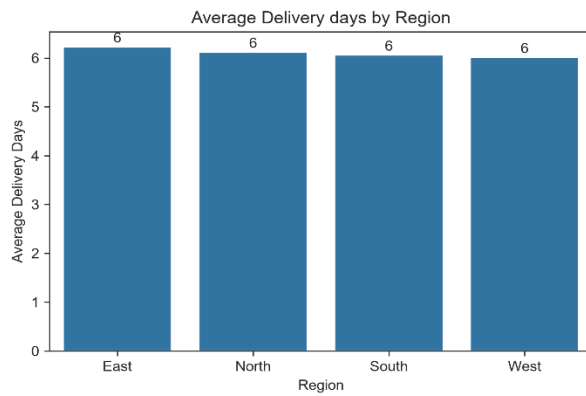
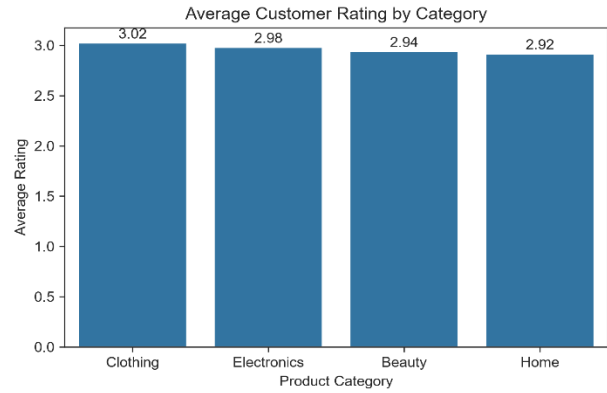
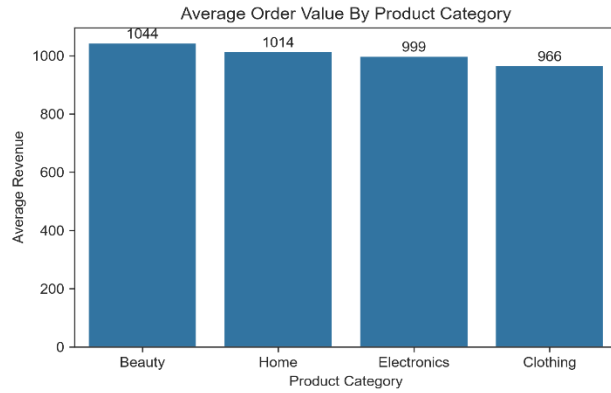
Monthly revenue, order volume, quantity sold and average discount were visualized to identify changes over time. Comparing these measures helps determine whether changes in revenue are accompanied by changes in transaction volume, quantity or discounting.

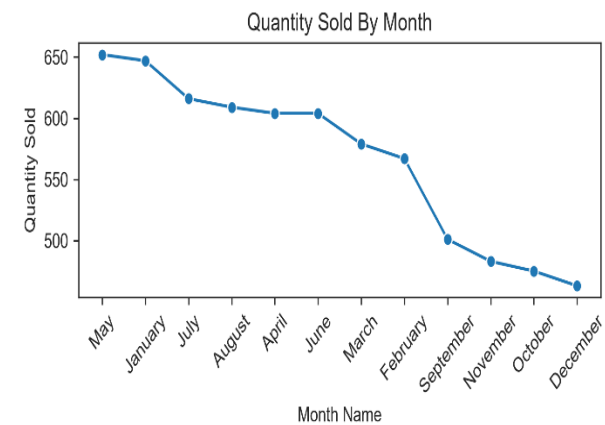
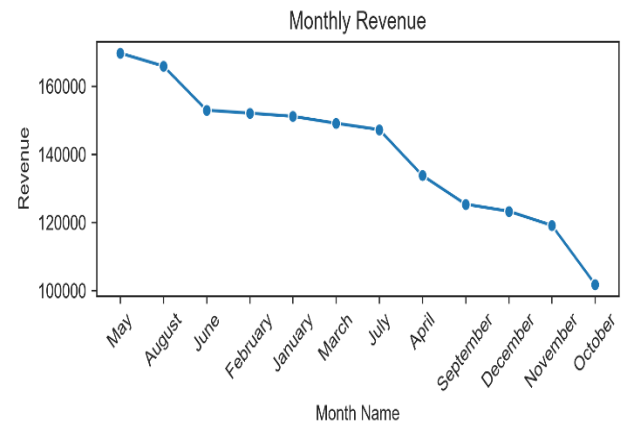
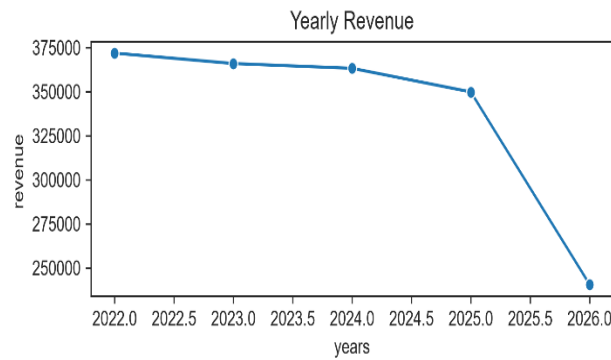
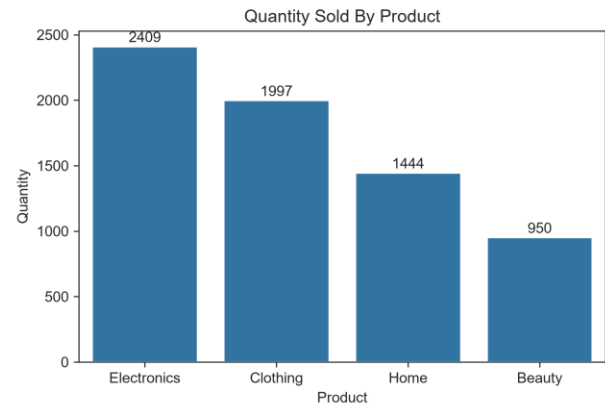
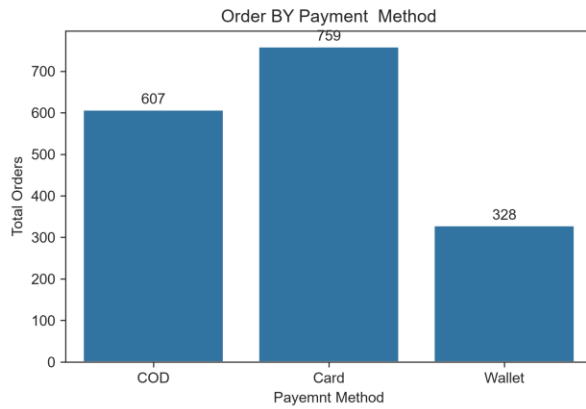
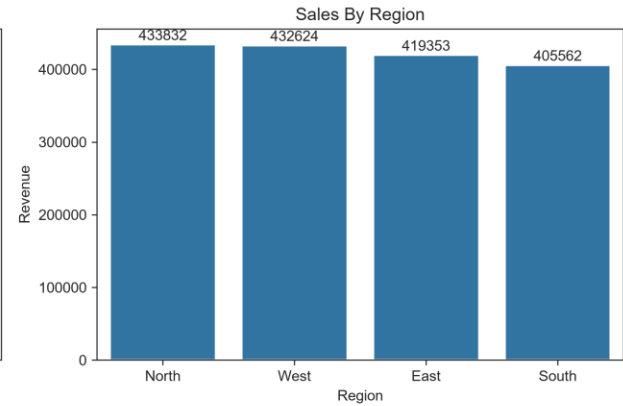
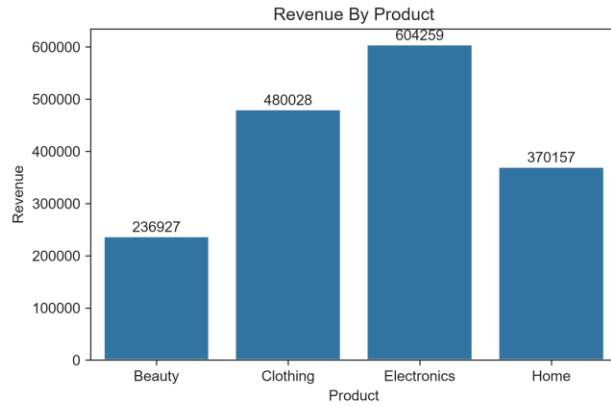
- High-performing months can be studied to identify successful products, campaigns or demand patterns.
- Low-performing periods can be used to plan targeted promotions, inventory and operational resources.
- Revenue trends should be interpreted together with orders and quantity rather than in isolation.

Delivery Performance

Average delivery days were compared across regions and delivery time was also examined against customer rating. This provides an operational perspective that complements the sales analysis.

- Regions with longer delivery times should be investigated for logistics bottlenecks.
- Improving delivery consistency may support better customer satisfaction.
- Delivery KPIs can be incorporated into regular regional performance monitoring.





Overall Conclusion

The analysis shows that e-commerce performance is influenced by multiple interconnected factors rather than a single variable. Revenue distribution, product category, quantity, discount, region, customer rating, payment method and delivery performance each provide a different perspective. The strongest business approach is therefore to combine these KPIs and use them together for decision-making.